Pratibimb

प्रतिबिंब (reflection)

A metabolic digital twin for Type 2 Diabetes that decides when to test

India-calibrated | Digital Twin Challenge 2026 | Happiest Health

Team SITARAM

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github.com/SriRamkunamsetty/SITARAM_QISCET

The problem: silent lows, and CGM is out of reach

~101 million

people with diabetes in India (ICMR-INDIAB-17, Lancet D&E 2023)

67% / 73%

In sulfonylurea-treated adults on CGM, lows were typically nocturnal (67%) and asymptomatic (73%) (Middleton, Diabetes Care 2017)

~Rs 9,000 / month

cost of continuous CGM in India: Rs 4,200-5,250 per 14-day sensor (2026 retail prices)

The question we answer

With only 1-4 test strips a day, WHEN should a patient test so that dangerous lows are predicted before they happen?

Our idea: a twin that learns, predicts, and decides when to test

1 Learn

A personal physiological model starts from the EHR and an Indian-specific prior, then learns the patient's insulin sensitivity, basal glucose and drug response from every reading.

2 Predict

Forecast glucose with honest uncertainty bands, and the probability of hypoglycaemia (below 70 and below 54 mg/dL) in the coming hours and overnight.

3 Decide

Choose the testing times that tell the twin the most about upcoming lows, and re-plan after every reading. Fewer strips, better warnings.

A classifier gives a number once. A digital twin keeps learning the patient and tells you what to do next.

Healthcare use case: one patient, one day

Enrol	Personalise	Plan	Warn	Audit
Doctor creates the twin from the EHR: age, BMI, waist, HbA1c, fasting glucose, regimen.	Indian endotype prior, then the twin learns from each reading plus the meal and dose log.	Each morning: the most informative testing times for the patient's strip budget.	At bedtime: overnight hypoglycaemia risk with an uncertainty band. Doctor reviews the dose.	Every number traces back to the exact readings and events behind it.

Who benefits

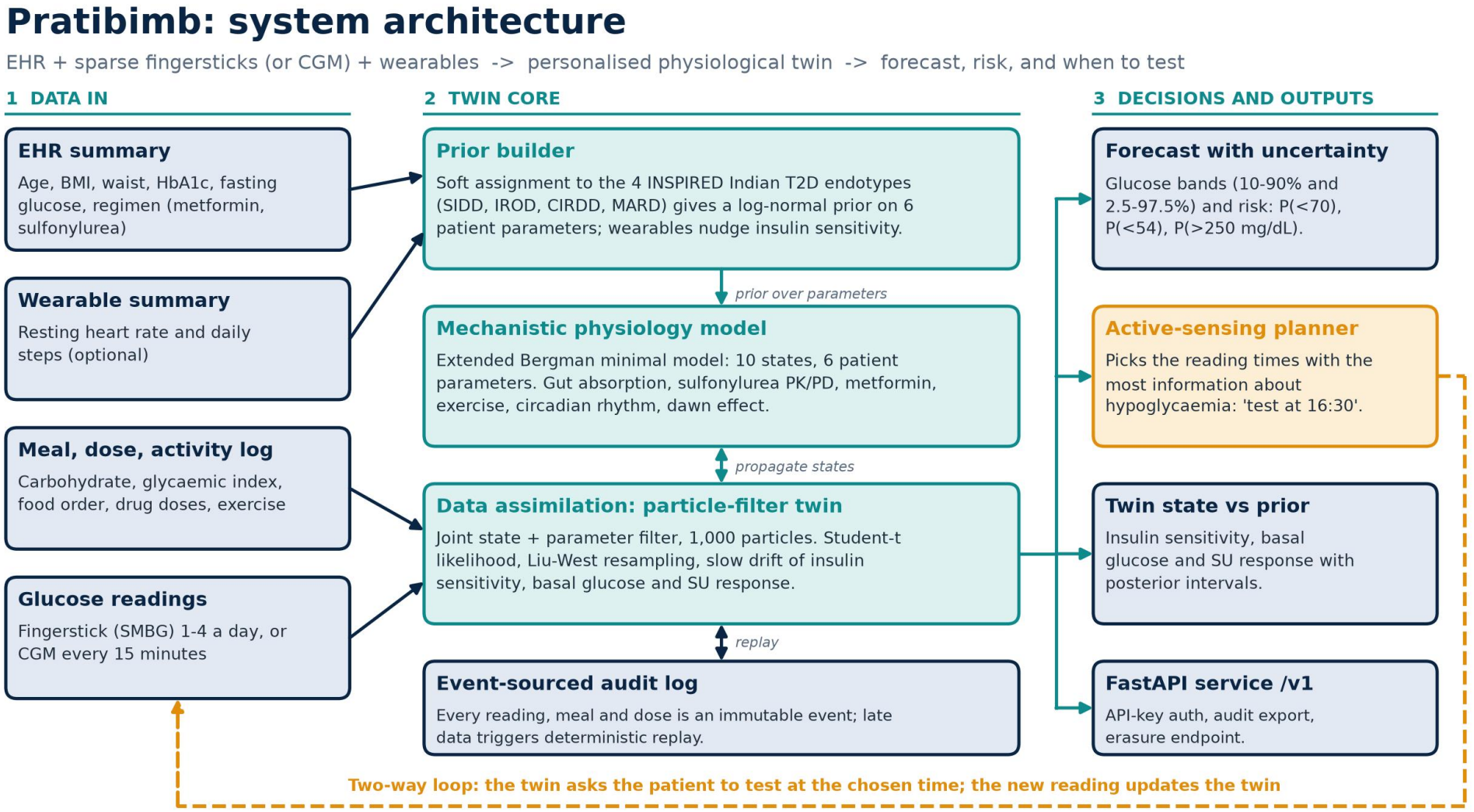
Patients on sulfonylureas without a CGM, and the diabetologists who must decide who needs review tonight, not next month.

Why this is a digital twin, not just a prediction model

NASEM (2023): a virtual representation that is dynamically updated from its physical twin, has predictive capability, and informs decisions.

NASEM element	Pratibimb
Virtual representation of a specific patient	10-state glucose, insulin, gut and drug model with 6 patient-specific parameters
Dynamically updated from the physical twin	Particle filter assimilates every reading (fingerstick or CGM)
Predictive capability	Posterior-predictive glucose trajectories and event probabilities
Informs decisions	Active-sensing planner (when to test), overnight risk at bedtime
Two-way data flow	The twin asks the patient to test at 16:30 and updates on the answer
Verification, validation, uncertainty	29 unit tests, known-truth evaluation, bootstrap CIs, full audit log

System architecture



AI / ML methods: hybrid physiology + probabilistic inference

Component	Method	Why
Mechanistic core	Extended Bergman minimal model: 10 states, 6 patient parameters, RK4 solver	Meaningful parameters; valid what-if simulation
Personalised prior	Soft assignment to 4 INSPIRED endotypes (n = 19,084), log-normal prior; wearables shift insulin sensitivity	Sensible twin on day 1, specific to Indian patients
Learning	Joint state + parameter particle filter, Student-t likelihood, Liu-West resampling, parameter drift	Learns online from sparse noisy data, with full posterior
Prediction	Posterior-predictive ensembles; P(<70), P(<54), P(>250 mg/dL)	Risk with uncertainty, not a point guess
Decision (active sensing)	Bayesian experimental design: greedy mutual information between readings and hypoglycaemia	Picks WHEN to test for the most information per strip

Stack: Python, NumPy (no GPU needed), FastAPI, Pydantic, Docker, GitHub Actions CI (Python 3.11 and 3.12).

How we tested it

Virtual Indian cohort with known ground truth

40 sulfonylurea-treated patients drawn from the INSPIRED endotype mix; 7 evaluation days each; 1,120 risk windows (morning, afternoon, evening, night).

Honest mismatch

The simulator differs from the twin: perturbed constants, daily parameter drift, carbohydrate-counting error, unlogged snacks, missed doses, device noise.

Fair comparison

Same truth, logs, meter noise and seed for every policy; patient-level paired bootstrap.

Testing policies compared

No readings: prior + logs only

Fixed: routine fasting / post-meal schedule

Random: random waking times

Twin-scheduled: information-maximising times, re-planned after every reading

CGM: reading every 15 minutes (reference)

Metric: AUROC and Brier score of P(low) at the start of each window.

Results: twin-chosen testing predicts lows better with fewer strips

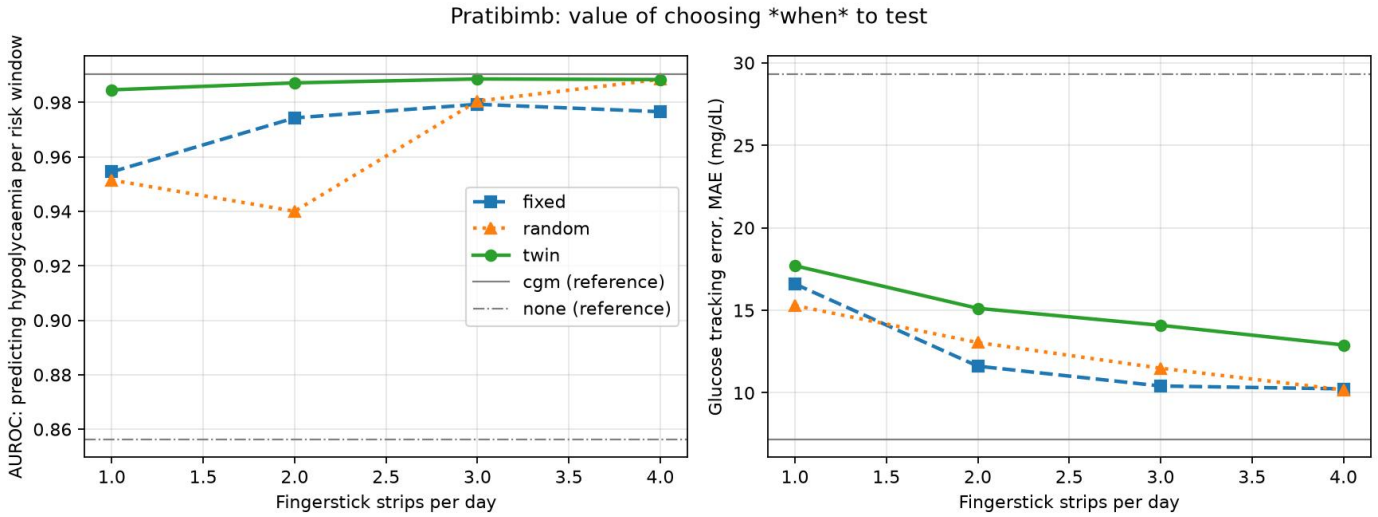


Chart: x-axis is the strip budget; the twin used fewer strips than the budget because it skips readings that would not change the risk estimate.

Policy	Strips/day	AUROC	Brier	Sens.
Fixed, budget 1	1.0	0.954	0.0251	0.72
Twin, budget 1	0.8	0.985	0.0167	0.79
Fixed, budget 2	2.0	0.974	0.0223	0.81
Twin, budget 2	1.4	0.987	0.0165	0.85
CGM (reference)	65.0	0.990	0.0153	0.81
No readings	0.0	0.856	0.0368	0.19

Brier improvement vs fixed schedule excludes zero at every budget (at 1 strip: +0.008, 95% CI +0.002 to +0.016).

AUROC / Brier: higher AUROC and lower Brier are better. Sens. = sensitivity at P >= 0.2.

Where it does not help

At 3-4 strips a day a random schedule performs about as well as the twin (no significant difference), and general glucose tracking is worse than the fixed schedule because the planner optimises for lows.

What we claim, and what we do not

What the evidence supports

At 1-2 strips a day, twin-chosen times predict hypoglycaemia better than the routine fasting / post-meal schedule.

The twin uses fewer strips: it skips readings that would not change the risk estimate.

With under one strip a day its risk ranking comes close to a CGM's (AUROC 0.985 vs 0.990).

Limitations we state openly

Only 5 of 40 virtual patients had any low (47 windows): small evidence base.

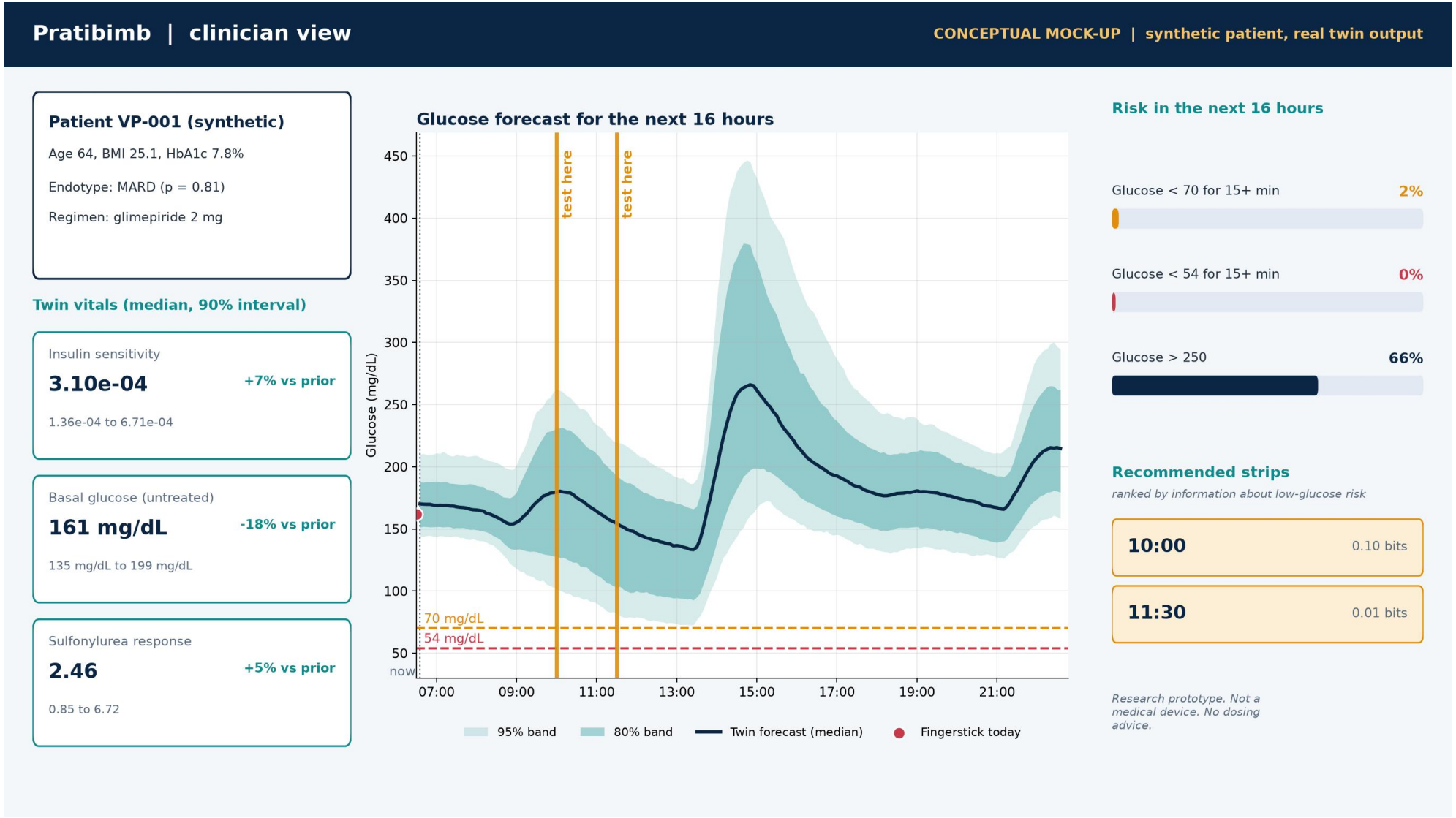
Synthetic data only; the twin and simulator share model structure. Real-data validation is next.

At 3-4 strips a random schedule matches the twin.

General glucose tracking is worse than the fixed schedule (14.1 vs 10.4 mg/dL at budget 3): the planner optimises for lows.

Bottom line: a promising result on synthetic patients. Validation on real patient data decides whether it holds.

Conceptual clinician dashboard



Safety, privacy and regulation by design

No dosing advice

Outputs are forecasts and risk. Every API response carries a research-prototype disclaimer.

Only synthetic and open data

No patient data in the repository; dataset licences documented in DATA.md.

Privacy by design (DPDP Act 2023 / Rules 2025)

No names or identifiers in the profile, erasure endpoint, full audit trail. Consent management is future work.

Regulatory awareness

CDSCO draft guidance on medical device software (21 Oct 2025) would treat a clinical version as SaMD; we make no clinical claims.

Auditability

Event-sourced: replaying the log reproduces every number the twin ever reported.

Next steps and how to run it

Roadmap

Validate on real open data: ShanghaiT2DM, then CGMacros

Conformal calibration of forecast bands

Interactive clinician dashboard (this mock-up is conceptual)

DPP-4 and SGLT2 inhibitor mechanisms; fasting-day (vrat / Ramadan) scenario

ABDM FHIR R4 import of the EHR profile

Run it

```
uv sync --all-extras
uv run pytest
uv run python scripts/demo_day.py
uv run uvicorn pratibimb.api.app:app
```

Team SITARAM

Apache-2.0

github.com/SriRamkunamsetty/SITARAM_QISCET

Thank you. Questions?